



Original Article

Prognostic importance of radiologic extranodal extension in nasopharyngeal carcinoma treated in a Canadian cohort



Olivia Chin^a, Eugene Yu^{a,b,1}, Brian O'Sullivan^{d,e}, Jie Su^c, Anais Tellier^a, Lillian Siu^f, John Waldron^{d,e}, John Kim^d, Aaron Hansen^f, Andrew Hope^d, John Cho^d, Meredith Giuliani^d, Jolie Ringash^{d,e}, Anna Spreafico^f, Scott Bratman^d, Ali Hosni^d, Ezra Hahn^d, Li Tong^d, Wei Xu^c, Shao Hui Huang^{d,e,1,*}

^a Department of Neuroradiology; ^b Department of Neuroradiology and Head and Neck Imaging, Princess Margaret Cancer Centre, University of Toronto; ^c Department of Biostatistics, Princess Margaret Cancer Centre/University of Toronto; ^d Department of Radiation Oncology; ^e Department of Otolaryngology – Head & Neck Surgery, Princess Margaret Cancer Centre, University of Toronto; and ^f Division of Medical Oncology, Princess Margaret Cancer Centre/University of Toronto, Canada

ARTICLE INFO

Article history:

Received 18 August 2021

Received in revised form 15 October 2021

Accepted 21 October 2021

Available online 27 October 2021

Keywords:

Magnetic resonance imaging

Nasopharyngeal carcinoma

Radiology

Staging

N classification

Extranodal extension

Survival

ABSTRACT

Purpose: To confirm the prognostic value of radiologic extranodal extension (rENE) and its role in clinical-N classification in nasopharyngeal carcinoma (NPC) treated in a western institution.

Methods and materials: NPC treated between 2010 and 2017 were included. Pre-treatment MRI were reviewed for unequivocal rENE and its grade: grade-1: tumour invading through any nodal capsule but confined to perinodal fat; grade-2: ≥ 2 adjacent nodes forming a coalescent nodal mass; grade-3: tumour extending beyond perinodal fat to invade/encase adjacent structures. Overall survival (OS) and disease-free survival (DFS) were compared between rENE-positive (rENE+) and rENE-negative (rENE-) patients. Multivariable analysis (MVA) confirmed the prognostic importance of rENE and its grade. Staging schemas including rENE in N-classification were proposed and their performance evaluated.

Results: A total of 274 patients were eligible (43 cN0; 231 cN-positive). rENE was identified in 83/231 (36%) cN-positive, including grade 1/2/3 rENE in 14/58/11 cases. Compared to rENE-, rENE+ patients had a lower OS (68% vs 89%, $p < 0.001$) and DFS (58% vs 80%, $p < 0.001$). MVA confirmed the prognostic importance of grade-2 [HR: OS: 2.85 ($p = 0.005$); DFS: 2.89 ($p < 0.001$)] and grade-3 rENE [HR: OS 5.28 ($p = 0.004$); DFS 3.86 ($p = 0.005$)], with a trend for grade-1 vs rENE- [HR: OS 2.63 ($p = 0.13$); DFS 1.49 ($p = 0.520$)]. We evaluated classifying any rENE as cN3 (Proposal-I) or any grade-2/grade-3 rENE as cN3 (Proposal-II). The stage schema with Proposal-I cN-classification ranked the highest in the performance evaluation.

Conclusions: rENE is an important prognostic factor in this western NPC cohort. We propose classifying any unequivocal rENE as cN3.

© 2021 Elsevier B.V. All rights reserved. Radiotherapy and Oncology 165 (2021) 94–102

Staging at the time of diagnosis is a crucial step in cancer management. It facilitates research activities, guides future treatment protocol development, and enhances cancer control activities [1]. The 8th edition TNM classification (TNM-8) unified nasopharyngeal cancer (NPC) staging systems among endemic and non-endemic jurisdictions and is used worldwide. Modification of T-classification criteria in TNM-8 reflected improved outcomes with contemporary NPC management based on datasets in NPC endemic regions [2]. Modification of cN-classification harmonized older terminology, employing “supraclavicular fossa” with recent consensus

nomenclature “lower neck lymph node” (i.e., levels IV and VB nodes) [3], and merged cN3a and cN3b as cN3.

TNM classification undergoes periodic updates to reflect improved understanding of tumour biology, and alteration of risk stratification profiles due to newer treatment options and prognostic factors. Unequivocal radiologic extranodal extension (rENE), an imaging representation of tumour invasion through a lymph node (LN) capsule, has emerged as a strong prognostic factor in NPC [4–7]. Previous studies [5–8] have classified extent of rENE into 3 grades, which are associated with worsening outcomes: grade-1 (G1): tumour invading through any LN capsule(s) but confined to perinodal fat; grade-2 (G2): a coalescent nodal mass involving ≥ 2 adjoining LNs; and grade-3 (G3): tumour extending beyond perinodal fat to invade adjacent structures. Given the prognostic importance of rENE in NPC, several authors [5–7] have proposed the inclusion of varying grades of rENE as a cN modifier. For exam-

* Corresponding author at: Department of Radiation Oncology, Princess Margaret Cancer Centre, 610 University Avenue, Toronto, ON M5G 2M, Canada.

E-mail address: shaohui.huang@rmp.uhn.ca (S.H. Huang).

¹ EY and SHH contributed equally.

ple, Lu et al. [6] proposed classifying any G3-rENE as cN3 category, TNM-8 cN1 with G2-rENE as cN2, and down-classify TNM-8 cN2 without rENE (rENE–) to cN1 disease. Mao et al. [5] proposed only including G3-rENE as a new cN modifier and classify any G3-rENE as cN3 disease. Both cN modification proposals showed improved survival distinction between different cN categories based on data from endemic regions. Whether this holds true in non-endemic populations and whether both aforementioned proposals are optimal remains uncertain.

NPC is mostly associated with Epstein-Barr viral (EBV) infection in endemic regions including Southeast Asia, East Asia, and North Africa [1]. In North America, the prevalence of NPC is relatively low, with a mixture of etiological factors including EBV (mostly amongst Asian ethnicity), HPV (typically in White European descendants), or traditional tobacco/alcohol consumption. Our institution is a tertiary referral cancer centre situated in an ethnically diverse metropolitan city where nearly 25% of the population is of Southeast Asian and Chinese descent, and 49% of European descent [9]. This provides an opportunity to confirm the prognostic value of rENE and its role in cN classification in an NPC cohort from a non-endemic region with diverse ethnicity, etiology, and histology.

Methods

Study population

Following institutional ethics board approval, we reviewed all newly diagnosed non-metastatic (M0) NPC treated with IMRT from January 2010 to December 2017. Cases for which pre-treatment gadolinium enhanced MRI was unavailable were excluded. Clinical information (demographics, disease, treatment, and outcomes) were retrieved from an in-house prospective *Anthology of Outcome System* where outcomes were recorded at point-of-care [10] and updated periodically from the provincial Cancer Registry. All patients were restaged according to the 8th edition TNM classification (TNM-8).

Scanning protocol

Our standard head and neck MRI scanning protocol includes the following sequences: a localizer sequence, sagittal, axial and coronal T1 pre-gad non-fat saturated, axial and coronal T2 with and without fat saturation, and axial and coronal T1 post-gad fat saturated sequences. The sequences were performed using a 20 × 20 cm field of view and with 3-mm slice thickness without slice gap. Gadolinium was administered unless contraindicated [11].

Radiologic nodal features

Two Neuroradiology fellows (AT with 1 year and OC with 2 years subspecialty experience) reviewed pre-treatment MRIs (within 10 weeks of IMRT commencement) of all cN-positive (cN+) NPC, blinded to clinical information and outcomes. Presence (based on unequivocal radiologic evidence) or absence (including negative, equivocal, or uncertainty) of the following nodal features were recorded: level, size, and side of radiologic overt LNs including retropharyngeal LNs (RPLN), lower neck involvement, and rENE. A radiologically overt LN was defined as nodal size >1.5 cm in largest dimension for level IIA, ≥8 mm for RPLN, and >1.0 cm for all other levels, or presence of suspicious features (e.g., necrosis, irregular borders, calcification, round shape) [11–14]. Lower neck LN referred to any involved LN with a border extending into or beyond level IV or VB per consensus definition [3]. rENE grade was also recorded based on published literature [5,8,12,15] (Fig. 1):

- G1-rENE: Tumour invading individual/separate lymph node(s) capsule(s) but confined to perinodal fat, characterized by clearly discernible loss of the sharp definition between the nodal capsule and its surrounding fat, or irregular nodal borders in individual LNs.
- G2-rENE: Tumour invading through two or more adjacent LNs which merge together to form a coalescent nodal mass characterized by partial or complete loss of the intervening planes (implying replacement by tumour).
- G3-rENE: Tumour extending beyond perinodal fat to overtly invade or encase adjacent structures, such as muscles, neurovascular structures, parotid, or skin. Encasement was defined as juxtaposed tumour encircling an anatomical structure by ≥270°.

To evaluate inter-rater reliability of rENE assessment and grading, 100 randomly selected cases were re-reviewed by a very experienced neuroradiologist (EY) (15 years of subspecialty experience). Kappa coefficient was calculated for inter-rater concordance.

Treatment and follow-up

All patients were managed in a multidisciplinary setting following institutional protocols (based on the 7th edition TNM) as described previously [16]. Generally, T1–T2 N0 patients received radiotherapy alone while T1–T2N1 were treated with concurrent chemoradiotherapy (if fit). More advanced cases were treated with concurrent chemoradiotherapy followed by adjuvant chemotherapy. The radiotherapy regimen comprised 70 Gy in 35 fractions over 7 weeks using IMRT with daily cone-beam for image guidance. Concurrent chemotherapy comprised cisplatin 100 mg/m² once every 3 weeks or 40 mg/m² weekly if comorbidities relevant to chemotherapy tolerance existed. Adjuvant chemotherapy comprised cisplatin 80 mg/m² on day 1 combined with 5-fluorouracil 1000 mg/m² in 24-hour infusion from day 2–5 every 28 days for a total of 3 cycles starting 4–6 weeks following the completion of chemo-radiotherapy.

Patients were followed in a multidisciplinary clinic and disease surveillance was conducted according to institutional protocol [17]. Dates of local or regional failure were recorded based on the date of histologic confirmation, while time of distant failure was recorded according to date of radiologic or indisputable clinical evidence.

Statistical analysis

Clinical characteristics and outcomes were compared between rENE-positive (rENE+) and rENE-negative (rENE–) cohorts. Chi-square tests (Fisher test for frequency <5) were performed for comparison of categorical variables, and Wilcoxon rank-sum tests for comparison of continuous variables. Overall survival (OS) (any cause of death as an “event”) and disease-free survival (DFS) (any local, regional, or distant failure, or death as an “event”) were estimated using the Kaplan–Meier method. Locoregional control (LRC) and distant control (DC) were calculated using the competing risk method where death without a specific failure event was considered as a competing-risk. All time-to-event occurrences were calculated from date of diagnosis. Log-rank tests and Gray's tests were used for outcome comparison. All tests were two-tailed with a probability of <0.05 considered statistically significant. Multivariable analyses (MVA) with Cox proportional hazard methods were applied to confirm the prognostic importance of rENE adjusting for age, TNM-8 T- and N-categories.

Finally, we investigated new cN-classification schemas (Supplement). The concordance indexes (C-index) (the higher the better)

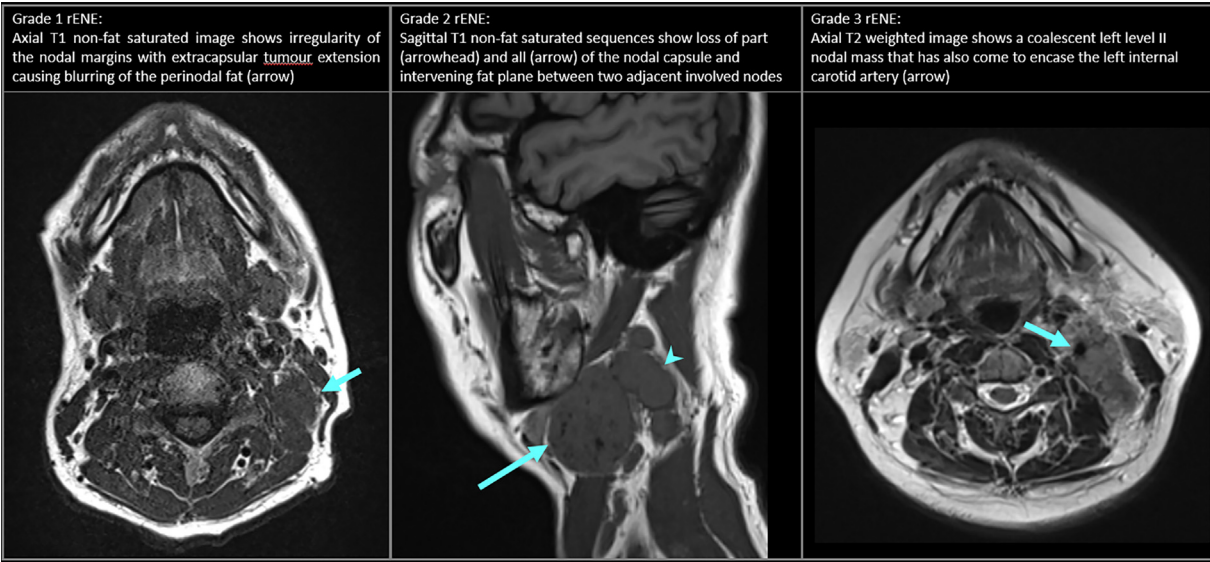


Fig. 1. MRI examples of Grade 1 (A), Grade 2 (B) and Grade 3 (C) radiologic extranodal extension. A. Grade-1 rENE: Axial T1 non-fat saturated image shows irregularity of the nodal margins with extracapsular tumour extension causing blurring of the perinodal fat (arrow). B. Grade-2 rENE: Sagittal T1 non-fat saturated sequences show loss of part (arrowhead) and all (arrow) of the nodal capsule and intervening fat plane between two adjacent involved lymph nodes, and C. Grade-3 rENE: Axial T2 weighted image shows a coalescent left level II nodal mass that has also come to encase the left internal carotid artery (arrow).

and the Akaike information criterion (AIC) (the lower the better) were calculated for various MVA models with different cN-classifications to compare predictability of prognosis and “goodness-of-fit”, respectively. The performance of OS and DFS prediction was evaluated in all cases (both cN0 and cN+) in the overall TNM staging using our cN classification proposal, as well as those proposed by Lu et al [6] and Mao et al [5], and compared with the current TNM-8 stage grouping schema based on well-established criteria using OS as the primary end-point and DFS as a secondary end-point [18–21]: (1). “Hazard consistency” (homogeneity): similar hazard among different T-N subgroups within each stage, (2). “Hazard discrimination” (distinctiveness): different hazard among different stages, and the higher the stage the higher the hazard, (3). “Explained variance”: the percentage of survival variation attributable to “stage” in the MVA, (4). “Likelihood difference”: the difference of ‘goodness-of-fit’ between the MVA models with and without the “stage” variable, and (5). “Sample size balance”: relatively balanced sample sizes across stage groups.

Results

A total of 285 consecutive patients were treated during the study period. Excluding 13 patients whose MRIs were unavailable, the remaining 274 patients (43 cN0 and 231 cN+) were eligible; their clinical characteristics are displayed in Table 1.

The entire cohort included 196 (72%) patients of Asian and 78 (28%) of non-Asian ethnicity. Etiological factors included 244 (89%) EBV-positive, 10 (4%) HPV-positive, 8 (3%) non-viral NPC, and 12 (4%) unknown aetiology. Histology included 6 (2%) keratinizing, 56 (20%) nonkeratinizing differentiated and 212 (77%) nonkeratinizing undifferentiated subtypes.

MRI features were reviewed for 231 cN+ NPC patients. Median interval from MRI acquisition to IMRT commencement was 2.1 weeks (0–10.4 weeks). TNM-8 cN categories were: 69 (30%) cN1, 125 (54%) cN2, and 37 (16%) cN3. Nodal disease was 146 (67%) unilateral and 47 (22%) bilateral. RPLN was present in 215 (93%) cases including 72 (33%) unilateral and 143 (67%) bilateral. Lower neck lymphadenopathy was present in 63 (27%) cases. The

majority of rENE (n = 83, 36%) was either G2-rENE (n = 58, 25%) or G3-rENE (n = 11, 5%) while G1-rENE only accounted for 14 (6%) of the cN+ cohort (Table 1). The frequency of rENE increased with higher cN category: 16% (11/69), 32% (40/125), and 86% (32/37) for the TNM-8 cN1, cN2, and cN3 subsets, respectively.

The interrater kappa for 100 randomly selected samples was 0.79 (95% CI: 0.66–0.91) for overall rENE dichotomized as “yes” vs “no”. For G1-rENE, G2-rENE, G3-rENE, and combined G2/3-rENE, the interrater kappas with 95% confidence interval (95% CI) were: 0.36 (0.04–0.69), 0.60 (0.42–0.77), 0.69 (0.54–0.85), and 0.69 (0.54–0.85), respectively (Supplement-1).

Median follow-up for cN+ was 5.7 years (range 0.6–11.3). Compared to rENE–, the rENE+ cohort had a lower 5-year OS [68% (95% CI 58–80) vs 89% (84–95) p < 0.001], DFS [58% (48–70) vs 80% (74–87) p < 0.001], and DC [67% (55–76) vs 89% (83–93) p < 0.001], but similar LRC [86% (76–92) vs 91% (85–95), p = 0.202]. Five-year OS declined with higher grade of rENE: 89% (84–95), 83% (64–100), 68% (56–83), and 55% (32–94) for rENE–, G1-rENE, G2-rENE, and G3-rENE, respectively. Five-year DFS rates were 80% (74–87), 84% (67–100), 54% (43–69), and 45% (24–87) for rENE–, G1-rENE, G2-rENE, and G3-rENE, respectively (p < 0.001) [Fig. 2].

MVA for OS, adjusted for age, TNM-8 T-category, and N-category, confirmed the prognostic importance of G2-rENE [HR 2.85 (1.36–5.95), p = 0.005] and G3-rENE [HR 5.28 (1.72–16.25), p = 0.004], and was marginally prognostic for G1-rENE [HR 2.63 (0.74–9.27), p = 0.13]. MVA for DFS confirmed the prognostic importance of G2-rENE [HR 2.89 (1.62–5.14), p < 0.001] and G3-rENE [HR 3.86 (1.50–9.92), p = 0.005], but was non-significant for G1-rENE [HR 1.49 (0.44–4.98), p = 0.520].

C-index and AIC for Lu [6] and Mao’s [5] proposal was improved slightly versus TNM-8 cN classification (c-index: 0.729 vs 0.706 vs 0.704; AIC: 535.60 vs 544.94 vs 545.30). To further improve OS prediction, we proposed two new cN classifications: Proposal-I classified any rENE+ as cN3 because HR for G1-rENE in MVA was still marginally prognostic despite its low frequency; Proposal-II classified only G2-rENE and G3-rENE as N3 because G1-rENE did not reach statistical significance in MVA and the inter-rater kappa was low. The c-index parameters for OS were further improved to 0.747 and 0.737 and AICs were also improved to 531.37 and 535.09

Table 1
Clinical characteristics of cN+ nasopharyngeal carcinoma patients.

Covariate	Total	cN0	cN+	cN+		p-value
				rENE–	rENE+	
Case Number	274	43 (16)	231 (84)	148 (64)	83 (36)	
Age [Median (range)] (years)		61 (17–88)	52 (18–80)	52 (18–80)	50 (22–78)	0.610
Gender						
Female	65	7 (16)	58 (25)	39 (26)	19 (23)	0.640
Male	209	36 (84)	173 (75)	109 (74)	64 (77)	
Ethnicity						
Asian	196	31 (72)	165 (71)	106 (72)	59 (71)	>0.999
Non-Asian	78	12 (28)	66 (29)	42 (28)	24 (29)	
ECOG PS						
ECOG 0	188	25 (58)	163 (71)	107 (72)	56 (67)	0.460
ECOG 1–2	86	18 (42)	68 (29)	41 (28)	27 (33)	
Smoking PY [Median (range)]		0 (0–50)	0 (0–90)	0 (0–90)	0 (0–50)	0.840
Primary Histology						
Keratinizing	6	3 (7)	3 (1)	3 (2)	0 (0)	0.013
Nonkeratinizing differentiated	56	8 (19)	48 (21)	38 (26)	10 (12)	
Nonkeratinizing undifferentiated	212	32 (74)	180 (78)	107 (72)	73 (88)	
Viral Association						
EBV	244	40 (93)	204 (93)	127 (91)	77 (98)	0.310
HPV	10	0 (0)	10 (4)	9 (6)	1 (1)	
Non-viral	8	3 (7)	5 (2)	4 (3)	1 (1)	
Unknown	12	0	12	8	4	
Treatment						
RT Alone	48	30 (70)	18 (8)	15 (10)	3 (4)	0.250
CCRT Alone	47	3 (7)	44 (19)	29 (20)	15 (18)	
CCRT-AC	174	10 (23)	164 (71)	100 (68)	64 (77)	
IC-CCRT	5	0 (0)	5 (2)	4 (3)	1 (1)	
rENE Grade						
No rENE	148	NA	148 (64)	148 (100)	0 (0)	<0.001
G1	14	NA	14 (6)	0 (0)	14 (17)	
G2	58	NA	58 (25)	0 (0)	58 (70)	
G3	11	NA	11 (5)	0 (0)	11 (13)	
Side(s) of Neck Disease						
Ipsilateral		NA	171	106	65	0.52
Bilateral		NA	47	33	14	
Retropharyngeal LN						
Ipsilateral		NA	72	47	25	0.31
Bilateral		NA	143	88	55	
Lower Neck LN(s)						
No		NA	168 (73)	145 (98)	79 (95)	0.250
Yes		NA	63 (27)	3 (2)	4 (5)	
Level VI LN(s)						
No		NA	224 (97)	145 (98)	79 (95)	0.250
Yes		NA	7 (3)	3 (2)	4 (5)	
TNM-8 T-category						
T1–T2	114	19 (44)	95 (41)	57 (39)	38 (46)	0.330
T3–T4	160	24	136 (59)	91 (61)	45 (54)	
TNM-8 N-category						
N1		NA	69 (30)	58 (39)	11 (13)	<0.001
N2		NA	125 (54)	85 (57)	40 (48)	
N3		NA	37 (16)	5 (3)	32 (39)	
Mao's N-category [6] (G3-rENE → N3)						
N1		NA	68 (29)	58 (39)	10 (12)	<0.001
N2		NA	123 (53)	85 (57)	38 (46)	
N3		NA	40 (17)	5 (3)	35 (42)	
Lu's N-category [5] (G3-rENE → N3; G2-rENE → one N stratum up; N2 without G2/3 rENE → N1)						
N1		NA	154 (67)	143 (97)	11 (13)	<0.001
N2		NA	8 (3)	0 (0)	8 (10)	
N3		NA	69 (30)	5 (3)	64 (77)	
Proposal I_N (Any rENE → N3)						
N1		NA	58 (25)	58 (39)	0 (0)	<0.001
N2		NA	85 (37)	85 (57)	0 (0)	
N3		NA	88 (38)	5 (3)	83 (100)	
Proposal II_N (Any G2/3 rENE → N3)						
N1		NA	60 (26)	58 (39)	2 (2)	<0.001
N2		NA	94 (41)	85 (57)	9 (11)	
N3		NA	77 (33)	5 (3)	72 (87)	

Abbreviations: rENE+: presence of radiologic extranodal extension; rENE–: absence of radiologic extranodal extension; ECOG PS: The Eastern Cooperative Oncology Group Performance Status Score; Smoking PY: smoking pack-years; RT: radiotherapy; CCRT: concurrent chemoradiotherapy; CCRT-AC: concurrent chemoradiotherapy followed by adjuvant chemotherapy; IC-CCRT: induction chemotherapy followed by concurrent chemoradiotherapy, G0: no radiologic extranodal extension; G1: grade 1; G2: grade 2; G3: grade 3. TNM-8: the 8th edition TNM classification.

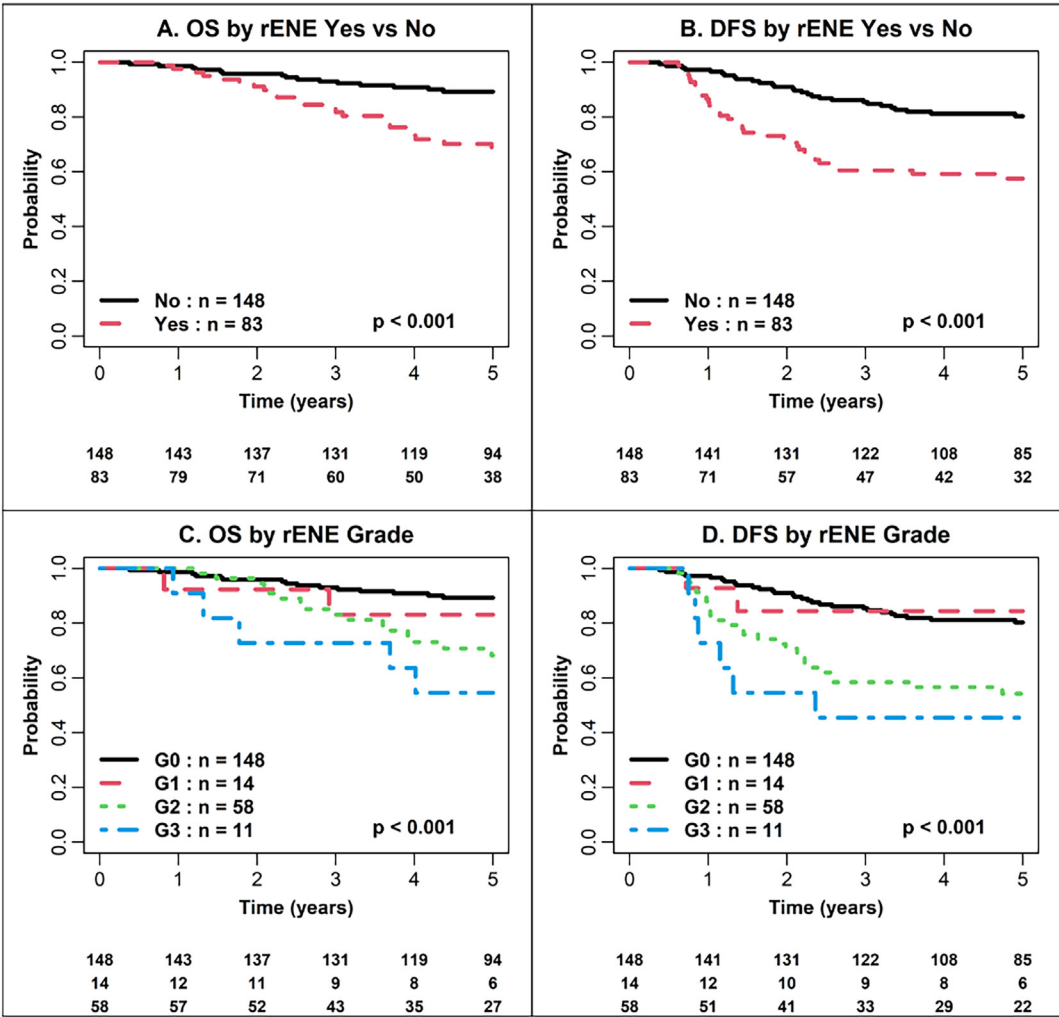


Fig. 2. OS and DFS by presence or absence of rENE (A, B) and rENE Grade (C, D).

for Proposal-I and Proposal-II, respectively. MVA results for various cN classification schema are detailed in Table 2. OS and DFS curves by cN for TNM-8, Mao, Lu, Proposal-I and Proposal-II are depicted in Fig. 3.

The performance of stage schema for OS and DFS was evaluated combining both cN0 and cN+ cases using the TNM-8 stage groupings. Stage schema using Proposal-I cN ranked first for both OS and DFS and Proposal-II second for OS and third for DFS (Table 3).

Discussion

Our study of NPC in a Western institution comprising 28% non-Asian descendants confirms the prognostic importance of rENE in addition to TNM-8 cN classification, and that its inclusion improves risk stratification. We propose two new cN classifications to relocate either any grades of rENE (Proposal-I) or only G2/G3-rENE (Proposal-II) to cN3 due to improved staging performance for OS and DFS prediction compared to the current TNM-8 cN-classification.

Similar to other studies [4-7,22,23], our study demonstrates that rENE is one of strongest prognostic factors and a good candidate for refining the current NPC cN classification. We found that rENE mainly increases risk of distant metastasis resulting in inferior DFS and OS, while LRC is less affected. The smaller impact on LRC likely reflects the radiosensitive nature of NPC: contemporary local treatment ablates tumour foci within irradiated volumes.

However, rENE increases the risk of tumour cell entering into blood stream where cannot be physically targeted or eradicated other than by potentially novel systemic strategies.

Besides strong prognostic value, TNM classification parameters must be reliably assessed. Our study demonstrates high concordance between radiologists in determining unequivocal rENE when dichotomized as “yes” vs “no”, regardless of experience as exemplified in this study between the observations of two Neuro-radiology fellows (AT with 1 year and OC with 2 years subspecialty experience) and a senior head and neck Radiologist with 15 years of subspecialty experience. However, inter-rater concordance is low for G1-rENE as shown in our study and others [5,6], which may be attributable to its low frequency and the technical challenges in ascertainment. rENE encompasses a continuous spectrum of nodal disease spread, from early breach of the LN capsule confined to the perinodal fat (likely reflected on imaging as G1-rENE) to invasion of adjacent anatomy, including the sternocleidomastoid muscle, parotid gland, neurovascular structures, and skin (depicted as G3-rENE on imaging). Radiological diagnosis in subtle examples of rENE may be subjective due to technical limitations such as motion artifact. Despite this, G1-rENE showed a trend towards lower OS after adjusting for TNM-8 T and N categories and age, with the HR approaching that of G2-rENE on MVA. This suggests that the prognostic value of G1-rENE may still be real in unequivocal cases.

In order to achieve widespread use and acceptance, a staging system should be simple and practical while still reflecting prognostic

Table 2
Multivariable analyses of OS and DFS for various cN classification proposals.

Multivariable Analysis	Overall Survival			Disease-free Survival		
	HR (95%CI)	p-value	Global p	HR (95%CI)	p-value	Global p
Variable						
rENE Grade			0.010			0.002
G1 vs G0	2.63 (0.74–9.27)	0.130		1.49 (0.44–4.98)	0.520	
G2 vs G0	2.85 (1.36–5.95)	0.005		2.89 (1.62–5.14)	<0.001	
G3 vs G0	5.28 (1.72–16.25)	0.004		3.86 (1.50–9.92)	0.005	
TNM-8 N-category			0.670			0.820
N2 vs N1	1.35 (0.62–2.95)	0.450		1.16 (0.63–2.13)	0.64	
N3 vs N1	1.03 (0.36–2.89)	0.960		1.28 (0.58–2.84)	0.54	
T-category: T3/4 vs T1/2	1.70 (0.90–3.23)		0.100	1.13 (0.68–1.89)		0.630
Age	1.07 (1.04–1.10)		<0.001	1.06 (1.03–1.08)		<0.001
MVA: TNM-8 cN						
TNM-8 N-category			0.016			0.019
N2 vs N1	1.72 (0.80–3.68)	0.160		1.33 (0.73–2.43)	0.350	
N3 vs N1	2.66 (1.11–6.39)	0.029		2.65 (1.33–5.27)	0.006	
T-category: T3/4 vs T1/2	1.70 (0.89–3.27)		0.110	1.10 (0.66–1.83)		0.710
Chemotherapy: Yes vs No	1.07 (0.37–3.07)		0.900	1.22 (0.50–2.94)		0.670
Age	1.07 (1.04–1.10)		<0.001	1.06 (1.03–1.08)		<0.001
C-index	0.704			0.673		
AIC	545.30			809.04		
MVA: Mao's cN						
Mao's N-category			0.073			0.019
N2 vs N1	1.65 (0.77–3.55)	0.200		1.31 (0.71–2.38)	0.390	
N3 vs N1	2.72 (1.15–6.43)	0.023		2.54 (1.29–5.02)	0.007	
T-category: T3/4 vs T1/2	1.72 (0.89–3.31)		0.100	1.10 (0.66–1.83)		0.710
Chemotherapy: Yes vs No	1.07 (0.37–3.06)		0.900	1.22 (0.50–2.94)		0.670
Age	1.07 (1.04–1.10)		0.001	1.06 (1.03–1.08)		<0.001
C-index	C-Index: 0.706			0.672		
AIC	544.94			809.26		
MVA: Lu's cN						
Lu's N-category			<0.001			<0.001
N2 vs N1	6.19 (1.83–20.98)	0.003		3.27 (1.15–9.32)	0.027	
N3 vs N1	2.83 (1.52–5.30)	0.001		2.99 (1.79–4.99)	<0.001	
T-category: T3/4 vs T1/2	2.11 (1.04–4.26)		0.037	1.23 (0.72–2.11)		0.450
Chemotherapy: Yes vs No	0.86 (0.30–2.46)		0.770	0.94 (0.39–2.31)		0.900
Age	1.06 (1.04–1.10)		<0.001	1.05 (1.03–1.08)		<0.001
C-index	0.729			0.708		
AIC	535.60			797.49		
MVA: Proposal-I cN						
Proposal-I_N (all rENE → N3)			<0.001			<0.001
N2 vs N1	2.39 (0.82–6.97)	0.110		1.28 (0.58–2.80)	0.540	
N3 vs N1	5.92 (2.22–15.83)	<0.001		3.47 (1.73–6.93)	<0.001	
T-category: T3/4 vs T1/2	2.07 (1.07–4.02)		0.031	1.27 (0.76–2.13)		0.370
Chemotherapy: Yes vs No	0.89 (0.31–2.55)		0.820	1.00 (0.42–2.42)		>0.999
Age	1.07 (1.04–1.10)		<0.001	1.06 (1.03–1.08)		<0.001
C-index for MVA	0.747			0.703		
AIC	531.37			796.86		
MVA: Proposal-II cN						
Proposal-II_N (G2/3 rENE → N3)			0.001			<0.001
N2 vs N1	2.13 (0.80–5.69)	0.130		1.22 (0.58–2.58)	0.590	
N3 vs N1	4.88 (1.93–12.32)	<0.001		3.41 (1.73–6.72)	<0.001	
T-category: T3/4 vs T1/2	1.96 (1.02–3.79)		<0.001	1.23 (0.73–2.05)		0.430
Chemotherapy: Yes vs No	0.84 (0.29–2.43)		0.750	0.93 (0.38–2.25)		0.860
Age	1.07 (1.04–1.10)		<0.001	1.05 (1.03–1.08)		<0.001
C-index	0.737			0.707		
AIC	535.09			797.35		

Abbreviations: C-index: the concordance index; AIC: the Akaike information criterion. OS: overall survival; DFS: disease-free survival; LRF: locoregional failure; DM: distant metastasis.

Note: The higher the C-index or the lower the AIC, the better the predictability of the statistical model. The model containing Proposal-I N has highest C-index for OS, and lowest AIC for both OS and DFS.

sis. rENE may be recorded in either a binary or graded manner. However, grading of rENE has not been universally accepted. Dichotomization of rENE is more practical for clinical use for now. Therefore, two cN classification schemas are proposed, both dichotomizing unequivocal rENE into “yes” vs “no”, the difference being inclusion of G1-rENE as “yes”. *Proposal-I* classifies all

unequivocal rENE as cN3, which appears to be the simplest approach while achieving the top overall ranking for both OS and DFS compared to other proposals. However, inclusion of G1-rENE leaves potential for the inexperienced radiologist to inadvertently “upstage” disease by “overcalling” equivocal cases as G1-rENE. Furthermore, G1-rENE refers to tumour confined to the perinodal

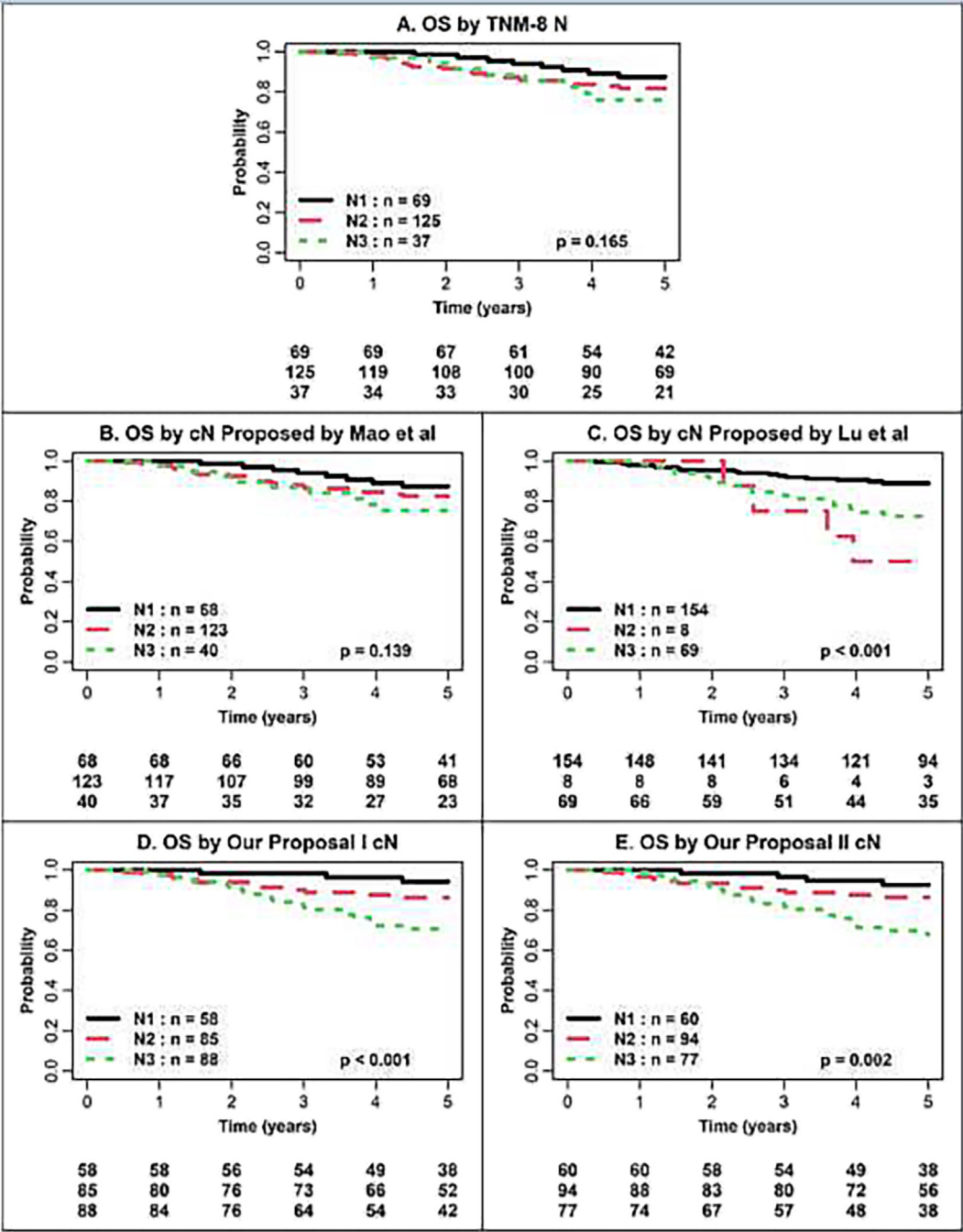


Fig. 3. OS by TNM-8 N (A), Mao's N (B), Lu's N (C), Proposal I-N (D) and Proposal II-N (E).

fat, which may be less prognostic if treated at this stage. Therefore, we also evaluated the performance of *Proposal-II* where only G2/G3-rENE were considered “yes” for rENE, but found it ranked lower than *Proposal-I*.

Others have also proposed incorporation of rENE in cN classification. Mao et al. [5] restricted only G3-rENE to cN3 and excluded G2-rENE due to non-significance for regional control despite significance for DM, DFS, and OS. However, our study showed that ignoring G2-rENE could reduce staging performance. Lu et al. [6] separated G2-rENE and G3-rENE in their proposal due to differential HRs associated with G2-rENE and G3-rENE. Essentially, cN1 disease with G2-rENE would be classified to cN2 not cN3. However, differentiating between grades of rENE in cN classification may add

an additional layer of uncertainty and complexity in practice, both of which are undesirable when a new staging parameter is introduced. Our *Proposal-I* and *Proposal-II* have the advantage of simplicity with better OS predictability (as shown in high c-index and low AIC). Both proposals outperformed TNM-8 and other staging schemas [5,6] that included rENE in cN classification for OS.

A unique feature of the current study is that the cohort is from a non-endemic region in a metropolitan jurisdiction in Canada, and as such comprising mixed ethnicity and aetiology. Although the majority are of Asian background, many represent first generation immigrants and their offspring. EBV is still the dominant etiologic factor for our study population, while 4% are caused by HPV and 2% due to non-viral (smoking) etiology. HPV + NPC is emerging as a

Table 3
Evaluation of performance in OS and DFS prediction with various cN classification proposals.

	TNM-8	Staging using Mao's N	Staging using Lu's N	Staging using Proposal I_N	Staging using Proposal II_N
OS					
Hazard Consistency	1.33	1.46	1.40	0.91	1.02
Score	0.76	1.00	0.88	0.00	0.19
Rank	3	5	4	1	2
Hazard Discrimination	1.23	1.19	0.17	0.23	0.45
Score	1.00	0.96	0.00	0.06	0.26
Rank	5	4	1	2	3
Explained Variance	12.05	11.90	15.02	16.92	15.78
Score	0.97	1.00	0.38	0.00	0.23
Rank	4	5	3	1	2
Likelihood Difference	4.35	4.14	6.46	8.81	7.11
Score	0.95	1.00	0.50	0.00	0.36
Rank	4	5	3	1	2
Balance	0.65	0.65	0.43	0.72	0.70
Score	0.77	0.77	0.00	1.00	0.95
Rank	2	3	1	5	4
Overall Score	4.07	4.34	1.76	0.56	1.52
Overall Rank	4	5	3	1	2
DFS					
Hazard Consistency	1.76	1.87	2.19	1.33	1.33
Score	0.51	0.63	1.00	0.01	0.00
Rank	3	4	5	2	1
Hazard Discrimination	1.26	1.09	0.81	1.86	1.78
Score	0.42	0.26	0.00	1.00	0.92
Rank	3	2	1	5	4
Explained Variance	11.58	11.35	14.79	15.71	15.09
Score	0.95	1.00	0.21	0.00	0.14
Rank	4	5	3	1	2
Likelihood Difference	5.38	4.98	8.66	10.90	9.76
Score	0.93	1.00	0.38	0.00	0.19
Rank	4	5	3	1	2
Balance	0.65	0.65	0.43	0.72	0.70
Score	0.77	0.77	0.00	1.00	0.95
Rank	2	3	1	5	4
Overall Score	3.19	3.27	1.59	1.51	1.72
Overall Rank	4	5	2	1	3

Note: all staging schemas used the same T-N combination according to the 8th edition TNM with various definition of N categories which were modification of current 8th edition N classification by inclusion of rENE parameter in various ways. Mao's proposal classified any grade 3 rENE as cN3. Lu's proposal classified graded 3 rENE as cN3, grade 2 rENE one N stratum higher, and N2 without rENE as N1. Our proposal-I classified all rENE regardless of grade as N3. Our proposal-II classified graded 2 and 3 rENE to N3 while grade 1 rENE was not included in the N classification.

Abbreviation: rENE–: no radiologic extranodal extension; rENE+: presence of extranodal extension.

new trend in western NPC populations, more often, though not exclusively, seen in people of European ancestry [24–26]. Whether HPV + NPC behaves differently from EBV + NPC remains to be elucidated.

This study has several limitations. Despite high inter-rater concordance for rENE, we do not have pathological extranodal extension data to confirm the accuracy of our rENE assessment, since this disease is uniformly treated with definitive (chemo-) radiotherapy. However, our data from oral cavity cancer have shown that unequivocal rENE has very high specificity for pENE [14]. The confirmed prognostic importance makes rENE a useful parameter for risk stratification and staging in NPC. EBV DNA was not included in the multivariable analysis model due to mixed etiology and unavailability in our cohort. However, it is unlikely to change the main finding of our study showing the prognostic importance of rENE in NPC. Furthermore, despite being one of the largest contemporary Western NPC cohorts, our sample size remains small with infrequent occurrence of G1-rENE. Since the frequency of G1-rENE were similar between Asian (12/165, 7%) and non-Asian (2/66, 3%) patients, collaborative datasets between different countries may elucidate whether G1-rENE is prognostically important for future TNM cN classification refinement. Finally, our proposals were based on a Canadian cohort with small sample size in an academic centre, and should be considered as a hypothesis-generating exercise. Whether it is reproducible in a

community setting and in endemic regions remains to be evaluated.

Conclusion

We confirm that unequivocal rENE has high inter-rater reliability and is prognostically important in NPC in a diverse Western cohort, almost a quarter of whom comprised patients of non-Asian ethnicity with 2% non-viral, 4% HPV aetiology. We agree with other authors' recommendation to include rENE as a cN modifier for future TNM classifications. Considering practicality and simplicity, we propose dichotomizing rENE as “yes” vs “no” and classifying any unequivocal rENE, either G1/G2/G3-rENE or G2/G3-rENE, as cN3. We report that inclusion of rENE in the cN classification can further improve risk stratification to facilitate segregation of patients into groups that may benefit from different treatment approaches or clinical trial entry in this non-endemic population. Validating our findings and proposals in larger independent datasets is warranted.

Funding

None.

Author's Contributions Statement

- Study concepts: Eugene Yu, Brian O'Sullivan, Shao Hui Huang
- Study design: Olivia Chin, Anais Tellier, Brian O'Sullivan, Eugene Yu, Shao Hui Huang,
- Data acquisition: All
- Quality control of data and algorithm: Olivia Chin, Anais Tellier, Eugene Yu, Shao Hui Huang, Jie Su, Li Tong
- Data analysis and interpretation: All
- Statistical Analysis: Jie Su, Wei Xu
- Manuscript drafting: Olivia Chin, Brian O'Sullivan, Shao Hui Huang
- Manuscript editing: all
- Manuscript review: all

Conflict of Interest

Drs. Giuliani, Hansen, Hope, Kim, and Spreafico declared non-financial support and/or advisory/consulting/research role outside the submitted work in their Conflict-of-Interest Disclosure Form.

No actual or potential conflicts of interest exist for any other authors on this topic.

Acknowledgements

We acknowledge the Bartley-Smith/Wharton, Gordon Tozer, Wharton Head and Neck Translational, Dr. Mariano Elia, Petersen-Turofsky, "The Joe & Cara Finley Center for Head & Neck Cancer Research," and "Discovery" Funds at the Princess Margaret Cancer Foundation for supporting the authors' (SHH, JS, WX, BOS) academic activities. We also acknowledge the O. Harold Warwick Prize of the Canadian Cancer Society for supporting the author's (BOS) academic activities.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.radonc.2021.10.018>.

References

- [1] Brierley J, O'Sullivan B, Asamura H, et al. Global Consultation on Cancer Staging: promoting consistent understanding and use. *Nat Rev Clin Oncol* 2019;16:763–71.
- [2] Pan JJ, Ng WT, Zong JF, et al. Proposal for the 8th edition of the AJCC/UICC staging system for nasopharyngeal cancer in the era of intensity-modulated radiotherapy. *Cancer* 2016;122:546–58.
- [3] Gregoire V, Ang K, Budach W, et al. Delineation of the neck node levels for head and neck tumors: a 2013 update. DAHANCA, EORTC, HKNPCSG, NCIC CTG, NCRI, RTOG, TROG consensus guidelines. *Radiother Oncol* 2014;110:172–81.
- [4] Karakurt Eryilmaz M, Kadiyran C. Prognostic significance of radiologic extranodal extension in nasopharyngeal cancer. *Otolaryngol Head Neck Surg* 2021. 1945998211008887.
- [5] Mao Y, Wang S, Lydiatt W, et al. Unambiguous advanced radiologic extranodal extension determined by MRI predicts worse outcomes in nasopharyngeal carcinoma: Potential improvement for future editions of N category systems. *Radiother Oncol* 2021;157:114–21.
- [6] Lu T, Hu Y, Xiao Y, et al. Prognostic value of radiologic extranodal extension and its potential role in future N classification for nasopharyngeal carcinoma. *Oral Oncol* 2019;99:104438.
- [7] Ai Q-Y, King AD, Poon DMC, et al. Extranodal extension is a criterion for poor outcome in patients with metastatic nodes from cancer of the nasopharynx. *Oral Oncol* 2019;88:124–30.
- [8] Hu Y, Lu T, Huang SH, et al. High-grade radiologic extra-nodal extension predicts distant metastasis in stage II nasopharyngeal carcinoma. *Head Neck* 2019;41:3317–27.
- [9] Statistics Canada. Census Profile, 2016 Census. 2016. <https://www12.statcan.gc.ca/census-recensement/2016/dp-pd/prof/index.cfm?Lang=E> (accessed July 20 2021).
- [10] Wong K, Huang SH, O'Sullivan B, et al. Point-of-care outcome assessment in the cancer clinic: audit of data quality. *Radiother Oncol* 2010;95:339–43.
- [11] Billfalk-Kelly A, Yu E, Su J, et al. Radiologic extranodal extension portends worse outcome in cN+ TNM-8 stage I human papillomavirus-mediated oropharyngeal cancer. *Int J Radiat Oncol Biol Phys* 2019;104:1017–27.
- [12] Huang SH, O'Sullivan B, Su J, et al. Prognostic importance of radiologic extranodal extension in HPV-positive oropharyngeal carcinoma and its potential role in refining TNM-8 cN-classification. *Radiother Oncol* 2020;144:13–22.
- [13] Sananmuang T, Yu E, Su J, et al. Pre- and POST-RADIOTHERAPY RADIOLOGIC NODAL FEATURES AND OROPHARYNGEAL CANCER OUTCOMES. *Laryngoscope* 2020; Accepted.
- [14] Almulla A, Noel CW, Lu L, et al. Radiologic-pathologic correlation of extranodal extension in patients with squamous cell carcinoma of the oral cavity: implications for future editions of the TNM classification. *Int J Radiat Oncol Biol Phys* 2018;102:698–708.
- [15] Huang SH, Chernock R, O'Sullivan B, Fakhry C. Assessment criteria and clinical implications of extranodal extension in head and neck cancer. *Am Soc Clin Oncol Educ Book* 2021:265–78. <https://doi.org/10.1200/JEBO.2020.39.320939>.
- [16] Oliva M, Huang SH, Taylor R, et al. Impact of cumulative cisplatin dose and adjuvant chemotherapy in locally-advanced nasopharyngeal carcinoma treated with definitive chemoradiotherapy. *Oral Oncol* 2020;105:104666. <https://doi.org/10.1016/j.oraloncology.2020.104666>.
- [17] Princess Margaret Head and Neck Site Group. Princess Margaret Cancer Centre Clinical Practice Guidelines: Head and Neck - Nasopharynx. Clinical Practice Guidelines. July, 2019 ed: Princess Margaret Cancer Program; 2019.
- [18] Huang SH, Xu W, Waldron J, et al. Refining American Joint Committee on Cancer/Union for International Cancer Control TNM stage and prognostic groups for human papillomavirus-related oropharyngeal carcinomas. *J Clin Oncol* 2015;33:836–45.
- [19] O'Sullivan B, Huang SH, Su J, et al. Development and validation of a staging system for HPV-related oropharyngeal cancer by the International Collaboration on Oropharyngeal cancer Network for Staging (ICON-S): a multicentre cohort study. *Lancet Oncol* 2016;17:440–51.
- [20] Xu W, Shen X, Su J, O'Sullivan B, Huang SH. Refining evaluation methodology on TNM stage system: assessment on HPV-related oropharyngeal cancer. *Austin Biom and Biostat* 2015;2:1014.
- [21] Groome PA, Schulze KM, Mackillop WJ, et al. A comparison of published head and neck stage groupings in carcinomas of the tonsillar region. *Cancer* 2001;92:1484–94.
- [22] Yin X, Lv L, Pan XB. Prognosis of extracapsular spread of cervical lymph node metastases in nasopharyngeal carcinoma. *Front Oncol* 2020;10:523956.
- [23] Tian YM, Zeng L, Lan YH. The value of cervical node features in predicting long-term survival of nasopharyngeal carcinoma in the intensity-modulated radiotherapy era. *Cancer Manag Res* 2021;13:4899–909.
- [24] Maxwell JH, Kumar B, Feng FY, et al. HPV-positive/p16-positive/EBV-negative nasopharyngeal carcinoma in white North Americans. *Head Neck* 2010;32:562–7.
- [25] Stenmark MH, McHugh JB, Schipper M, et al. Nonendemic HPV-positive nasopharyngeal carcinoma: association with poor prognosis. *Int J Radiat Oncol Biol Phys* 2014;88:580–8.
- [26] Wotman M, Oh EJ, Ahn S, Kraus D, Costantino P, Tham T. HPV status in patients with nasopharyngeal carcinoma in the United States: A SEER database study. *Am J Otolaryngol* 2019;40:705–10.